2D CFD Analysis and Aerodynamic Validation of the Flow Past a Square Cylinder at $Re = 13,000$

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Abstract

This technical report presents an Unsteady Reynolds-Averaged Navier-Stokes (URANS) numerical analysis of the flow past a square cylinder at a Reynolds number of $Re = 13,000$ in the ANSYS Fluent environment. Utilizing the SST $k-\omega$ turbulence model, special emphasis is placed on near-wall mesh resolution (through the y^+ parameter) and temporal discretization to accurately capture the characteristic phenomenon of this case: the vortex shedding. The computed parameters (mean aerodynamic drag coefficient, $\overline{C_d}$, and the Strouhal number, St) are validated against the experimental benchmark carried out by Norberg in 1993 [1]. In such study, different tests related to the flow and pressure forces were carried out over various rectangular cylinders with variations of their angle of attack ($\alpha = 0 - 90°$) and their sides' legth ratio ($A/B = 1; 1.62; 2.5; 3$).

1 Introduction

The flow around a square cylinder is a canonical benchmark in Computational Fluid Dynamics (CFD) due to its fixed separation points at the sharp leading edges and the subsequent complex vortex shedding dynamics. Unlike circular cylinders, where the separation location oscillates due its geometry, the sharp corners of a square cylinder anchor the separation point, making it an ideal test case for validating turbulence models such as the SST $k-\omega$ formulation.

This document outlines the numerical setup, boundary conditions, near-wall treatment and post-processing methodologies implemented to reproduce experimental results for a cylinder of side $L = 0.01$ m at $Re = 13,000$ for $\alpha = 0°$ and $A/B = 1$.

2 Numerical Setup and Methodology

2.1 Governing Equations and Turbulence Model

The simulations were conducted using the Navier-Stokes equations for unsteady and incompressible flow. Such equations are listed below, where Equation 1 represents the continuity equation and Equation 2 gathers the momentum equations for the 3 components of the reference frame (Cartesian in this case):

$$\frac{\partial U_j}{\partial x_j} = 0 \tag{1}$$

$$\frac{\partial U_j}{\partial t} + U_j \frac{\partial U_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \nu\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) \tag{2}$$

In order to achieve the turbulence closure for the aforementioned equations, the SST $k-\omega$ model was chosen due to its well-known superior performance in adverse pressure gradients and boundary layer separation regions.

This model, developed by Menter in 1993, combines the formulation of the $k-\omega$ model in the inner parts of the boundary layer (from the wall to the viscous sublayer) with the behavior of the $k-\varepsilon$ model in the freestream. Both transport equations, one for the turbulent kinetic energy k and one for the specific turbulent dissipation rate ω, are shown in Equations 3 and 4 respectively.

$$\frac{\partial k}{\partial t} + U_j \frac{\partial k}{\partial x_j} = P_k - \beta^* k\omega + \frac{\partial}{\partial x_j}\left[(\nu + \sigma_k \nu_T)\frac{\partial k}{\partial x_j}\right] \tag{3}$$

$$\frac{\partial \omega}{\partial t} + U_j \frac{\partial \omega}{\partial x_j} = \alpha S^2 - \beta\omega^2 + \frac{\partial}{\partial x_j}\left[(\nu + \sigma_\omega \nu_T)\frac{\partial \omega}{\partial x_j}\right] + \dots \\ \dots + 2(1 - F_1)\sigma_{\omega 2}\frac{1}{\omega}\frac{\partial k}{\partial x_i}\frac{\partial \omega}{\partial x_i} \tag{4}$$

Where F_1 es known as the bending function and is what allows CFD algorithms to switch between both pre-existing turbulence models: $F_1 = 0$ for $k-\varepsilon$ and $F_1 = 1$ for $k-\omega$ (the definition of F_1, alongside the corresponding definitions of the several coefficients present in the SST $k-\omega$ formulation can be consulted on CFD Online's webpage).

2.2 Computational Domain and Boundary Conditions

The control volume consisted of a velocity inlet located $5L$ upstream, a pressure outlet $15L$ downstream and symmetric side walls positioned $5L$ from the centerline, as shown in Figure 1.

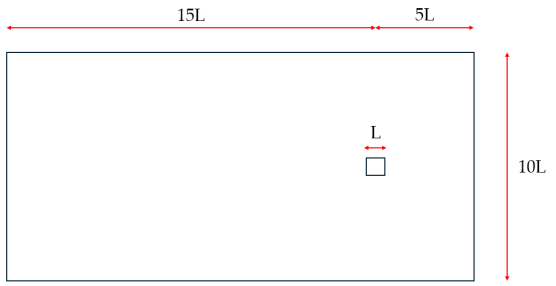


Figure 1: Computational domain

Related to the inlet's velocity condition, aiming to acquire the desired Reynolds number, a velocity magnitude of 19.1 m/s was established, since ISA conditions for air density ($\rho = 1.225$ kg/m^3) and dynamic viscosity ($\mu = 1.8 \cdot 10^{-5}$ Pa $\cdot$ s) were considered. Additionally, a turbulence intesity of 5% was introduced.

On the other hand, the gauge pressure of the outlet was set to 0 Pa, which is equivalent to work with an absolute pressure of $101,325$ Pa (Note: $p_{abs} = p_{op} + p_{gauge}$).

Lastly, both stationary wall and no-slip conditions were applied to the cylinder itself.

2.3 Mesh Resolution

To properly capture the vortex shedding phenomenon, a family of 4 meshes, from M01 to M08, was developed. Table 1 includes both the number of elements and nodes for each mesh and Figure 2 serves as an illustrative example for the M01 mesh.

Table 1: Mesh family

	M01	**M02**	**M04**	**M08**
$N_{elements}$	31200	61980	124800	250240
N_{nodes}	31670	62644	125740	251571

For a subsequent mesh sensitivity analysis, the following refinement criterion was applied to each edge of the computational dominion: let N_i be the number of elements of any edge before the refinement and N_{i+1} after such process, the refinement factor is $r_{i,i+1} = N_{i+1}/N_i = 2^{1/D}$, where D is the dominion's dimensionality ($D = 2$ for this case). Nevertheless, since this refinement factor is used for 2 different directions, X and Y, the total refinement factor, r_t is of order 2: $r_t \sim r_X \cdot r_Y \sim 2^{1/D} \cdot 2^{1/D} \sim 2^{2/D} \sim 2$.

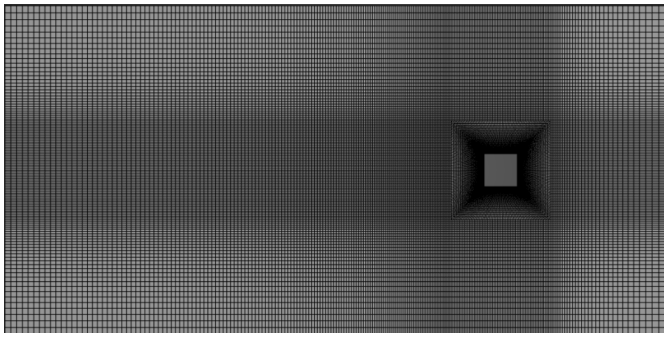


Figure 2: M01

3 Results and Discussion

Note: before addressing the results of the different simulations, in the Report Definitions section, due to the unusual position of both inlet and outlet for this case, the components of the Force Vector for the drag coefficient must be $(-1, 1)$.

The selected solving algorithm was the SIMPLE one (which stands for Semi-Implicit Method for Pressure Linked Equations). This algorithm takes the divergence of Equation 6 and substitues such result into Equation 5 in order to obtain an expression for the pressure gradient ∇p.

$$\nabla \cdot \mathbf{U} = 0 \tag{5}$$

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot (\mathbf{U}\mathbf{U}) - \nabla \cdot (\nu \nabla \mathbf{U}) = -\nabla p \tag{6}$$

The momentum equation can be rewritten in a semi-discretized way:

$$a_p \mathbf{U}_p = -\sum_n a_n \mathbf{U}_n + \frac{\mathbf{U}^0}{\Delta t} - \nabla p = H(\mathbf{U}) - \nabla p \tag{7}$$

Where:

- a_p represents any physical phenomenons that are discretized for the p cell of the computational dominion.
- a_n gathers any physical phenomenons discretized for the neighboring n cells of the p cell.

- $\mathbf{U}^0/\Delta t$ is the non-stationary and source term.

In the same way, the continuity equation can be discretized:

$$\nabla \cdot \mathbf{U} = \sum_f \mathbf{n} \cdot \mathbf{U}_f = 0 \Leftrightarrow \mathbf{U}_f = \frac{H(\mathbf{U})}{a_p}\bigg|_f - \frac{\nabla p}{a_p}\bigg|_f \quad (8)$$

Where:

- $\mathbf{n}$ is the normal vector of any face f.
- $\mathbf{U}_f$ is the velocity of face f.

Thus:

$$\nabla \cdot \left[\frac{H(\mathbf{U})}{a_p} - \frac{\nabla p}{a_p}\right] = 0 \Leftrightarrow \frac{\nabla^2 p}{a_p} = \nabla \cdot \left[\frac{H(\mathbf{U})}{a_p}\right] \quad (9)$$

As the reader might notice, Equation 9 happens to be Poisson's equation for the pressure p.

The time-history of the $\overline{C_d}$ was monitored for each of the meshes. However, only in M04 and M08 this coefficient reached a statistically stationary periodic state, while for M01 and M02 it acquired constant values.

Once the simulations reached the convergency levels, the $\overline{C_d}$ values were compared to Richardson's extrapolation, a method for improving numerical results whose main assumption is that the result of a discretised calculation can be expressed in terms of the results from the solutions of the simulations themselves (Equation 10). The main goal of this comparison is estimate an "exact" solution and to quantify the discretization error.

$$f_{\text{Richardson}} = f_1 + \frac{f_1 - f_2}{r^p - 1} \quad (10)$$

Where f_1 and f_2 are the solutions of the fine and coarse meshes (M08 and M04, respectively), r is the already mentioned refinement factor and p is the order of the numerical method ($p = 2$ in this case). Also, a Richardson's error, ε_R, can be defined and is shown in Equation 11.

$$\varepsilon_R = 100\frac{|f_{\text{Richardson}} - f|}{f_{\text{Richardson}}} \quad (11)$$

Being f the solution of the treated mesh. Table 2 collects the numerical values of $\overline{C_d}$ and ε_R, as well as including a new error, ε_N, which is calculated with the same method as Richardson's error (substituting $f_{\text{Richardson}}$ with Norberg's experimental value of $C_d = 2.16$).

Observing these results, the mesh sensitivity analysis indicates that the mesh that shows a balanced trade-off between numerical precision and simulation requirements, as well as acceptable y^+ values, was M04.

Finally, by exploiting the oscillatory nature of the $\overline{C_d}$ curves for M04 and M08 meshes, the vortex shedding frequency, f_s, can be used alongside the cylinder's side L and the inlet's velocity U_∞ to calculate the Strouhal number, St. This dimensionless number, expressed in Equation 12, represents the ratio of inertial forces due to the local acceleration of the flow to the inertial forces due to its convective acceleration [2].

Table 2: Numerical results

Parameter	M01	M02	M04	M08
$\overline{C_d}$	1.73	1.77	2.31	2.32
ε_R [%]	25.47	24	0.49	0.24
ε_N [%]	19.91	18.06	6.94	7.41

$$St = \frac{f_s L}{U_\infty} \quad (12)$$

For both M04 and M08, f_s was approximately 263.2 Hz, which resulted in a Strouhal number of 0.138. Compared to Norberg's experimental value, $St = 0.132$, the numerical result showed an error of $\varepsilon_N = 4.55$ %.

4 Conclusions

- **Mesh behavior and resolution.** The coarser meshes, M01 and M02, were insufficient to capture the transient and oscillatory nature of vortex shedding, yielding steady-state $\overline{C_d}$ values with errors of more than 18% compared with the experimental data.
- **Aerodynamic validation and Strouhal number.** The $\overline{C_d}$ analysis for M04 and M08 showed a shedding frequency of 263.2 Hz, yielding a Strouhal number of 0.138 and a deviation of 4.55%.
- Should the reader be interest in further study of this project, this link will redirect to a GitHub repository that contains this report and the necessary files for the simulations.

References

[1] Norberg, C. (1993). *Flow Around Rectangular Cylinders: Pressure Forces and Wake Frequencies*. Journal of Wind Engineering and Industrial Aerodynamics, Vol. 49, pp. 187-196.

[2] Katapodes, N. (2018). *Free Surface Flow: Computational Methods*, Ch. 5.10.3.